An Island Remote Sensing Image Segmentation Algorithm Based on FC_U-Net Network

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***ABSTRACT*-With the increasing importance of islands in many fields, it has become the focus of research to obtain information from island remote sensing images efficiently by using image semantic segmentation algorithm. In recent years, deep learning methods based on convolutional neural network have been widely used in image segmentation. However, for remote sensing images with rich details and different target scales, there are still some problems such as insufficient segmentation accuracy. To handle these problems, we propose an island remote sensing image segmentation algorithm based on FC_U-Net network. In this network, U-Net and FCN (Fully Convolutional Network) are improved respectively to form two new modules, and then the semantic features and detail features from the two modules are fused by a fusion module. The testing and comparative experiments on NWPU-RESISC45 dataset show that the quantitative metrics and visual effects of FC_U-Net are better than other three state-of-the-art methods, U-Net, FCN (Fully Convolutional Network), SegNet, which indicates the effectiveness of our method.**

Keywords-Remote sensing image; Island; Semantic segmentation; FC_U-Net network; Features fusion.

I. INTRODUCTION

Marine resources play a more and more important role in today's world. As the main carrier for the utilization and development of Marine, the status of island is also increasingly important. A large area of feature information contained in remote sensing images provides a data basis for the study of island[1]. Image semantic segmentation refers to the classification of each pixel in the image. That obtain information from remote sensing images efficiently by using image semantic segmentation algorithm has increasingly become the focus of scholars' research.

In recent years, scholars at home and abroad have done a lot of research in the field of image semantic segmentation and put forward many segmentation methods. In particular, the methods based on deep convolutional neural network (CNN) [2] have been rapidly developed and widely applied in the field of computer vision. Most of these methods based on CNN apply downsampling-to-upsampling network architectures to extract the high-layer semantic features, which can circumvent background disturbance and well segment large islands. However, due to the downsampling operations, these methods

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lack detail information and are not suitable for processing very small islands.

Based on the CNN method, many segmentation methods combining semantic features and detail features are proposed. In 2015, Long et al.[3] proposed FCN (Fully Convolutional Network), which replaced the full connection layer of CNN with the convolutional layer, realized the semantic segmentation of images in the form of end-to-end for the first time, and used skip structure to connect the shallow detail features and the deep semantic features. In 2015, Ronneberger et al.[4] proposed U-Net, which transferred all features from the encoder to the corresponding decoder and connected them to obtain detail and semantic features. In addition, with only a few tagged images, U-Net can obtain a high segmentation accuracy, and has achieved very good performance in biomedical segmentation applications. In 2015, Badrinarayanan et al.[5] proposed SegNet, in which decoder used the pooling indexes provided by the corresponding encoder to perform nonlinear upsampling operations, saving storage space and time. In 2016, Chen, Liang-Chieh et al.[6] proposed DeepLab, which segmented images with deep convolutional nets, atrous convolution, and fully connected CRFs. In 2017, Yu, Fisher et al.[7] proposed Deep Layer Aggregation, in which two structures of Deep Layer Aggregation (DLA) were proposed: Iterative Deep Aggregation (IDA) and Hierarchical Deep Aggregation (HDA). IDA was focused on merging different resolutions and scales, while HDA was focused on merging features from all modules and channels.

These segmentation methods have achieved good results in natural scene image segmentation. However, compared with natural scene images, remote sensing images often contain richer scale information and complex background interference. Therefore, in the process of segmentation, some small and medium-sized targets may be missed, and the target boundary cannot be segmented accurately. To handle these problems, we put forward FC_U-Net network for remote sensing image segmentation. The network structure includes three modules, a module based on U-Net network (Module A), a module based on FCN network (Module B), and another module to incorporate two modules (Module C). This network can extract not only semantic features, but also detail features, which greatly improve the segmentation performance. The main innovations are as follows:

(1) Deepen the depth of U-Net to extract more complex image features.

(2) The BN (Batch Normalization) layers[8] are added to normalize the features from the previous layers, which make the features more evenly distributed and improve the fault tolerance of the model while accelerating the convergence speed of the model.

(3) Dropout function[9] is used to randomly delete the neuron nodes in the hidden layers to reduce network complexity, reduce overfitting and enhance network generalization ability.

(4) A module based on FCN network (Module B) is added, and the downsampling layers are removed to avoid losing detail information.

(5) Integrate the detail features from Module B (the module based on FCN network) and semantic features from Module A (the module based on U-Net network) to improve the segmentation performance.

II. THE NETWORK STRUCTURE OF FC_U-Net

The network structure of FC_U-Net is shown in figure 1. Module A extracts semantic features, Module B extracts detail features, and module C fuses all features for segmentation.

A. The Network Structure of Module A

Module A is proposed based on U-Net. U-Net network is a deformation of the convolutional neural network. The encoder reduces the spatial resolution of the output features gradually through convolutional and pooling operations, while the decoder repairs the details and spatial resolution gradually through upsampling operations. There are some skip connections between the encoder and the decoder, thus helping the decoder repair the details better. However, the depth and width of the classical U-Net network are insufficient so that the feature information of different scales contained in the image cannot be fully learned. Therefore, we deepen the network layers (from A16 to A19 and from A22 to A24), so as to extract feature information more fully. In addition, we add BN (Batch Normalization) layers (A2, A6, A10, A14, A18) to normalize the features from the previous layers, which make the features more evenly distributed and improve the fault tolerance of the model while accelerating the convergence speed of the model. In addition, we add the Dropout function with rate=0.5 after the last encoding layer (A21), which randomly deletes the neuron nodes in the hidden layers to reduce network complexity, reduce overfitting, and enhance network generalization ability. Finally, Module A outputs a wealth of image semantic features with a small amount of detail features.

B. The Network Structure of Module B

In order to extract more detail information, Module B based on FCN is proposed. In FCN, the encoding layers convolve and pool the image, the decoding layers upsample the image, the shallow layers and the deep layers are connected by the skip structure. However, the downsampling operations of pooling layers will cause information loss, and the upsampling operations cannot recover the lost information losslessly. So Module B removes the pooling layers to prevent information loss and save image details. However, the removal of pooling layers leads to the decrease of the receptive field[10], which will reduce the prediction accuracy of the whole model. In order to solve this problem, we deepen the convolutional layers to expand the receptive field, reduce background interference, and improve the prediction accuracy. Finally, Module B outputs a wealth of detail features with less background interference.

C. The Network Structure of Module C

Module C combines the semantic features from Module A and the detail features from Module B. As shown in figure 1, Module C consists of 1 connection layer (C1) and 5 convolution layers (from C2 to C6). C1 combines the features of Module A and Module B through concatenation operation. Then, C2~C5 further extract features from the merged features. Finally, C6 calculates the fractional graph of image segmentation.

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III. EXPERIMENT AND ANALYSIS

A. Dataset Preparation

Remote sensing image data of this experiment is derived from NWPU-RESISC45[11]. This dataset is a publicly available benchmark for Remote Sensing Image Scene Classification (RESISC), created by Northwestern Polytechnical University (NWPU). It contains 31,500 images, covering 45 scene classes with 700 images in each class. These 45 scene classes include airplane, airport, baseball diamond and so on. We select island as the dataset of this experiment, with a total of 700 images whose pixel size is 256*256. The images are marked by ourselves. All images are randomly divided into a training dataset, a validation dataset and a testing dataset according to the ratio of 8:1:1. The training dataset includes 560 images, the validation dataset includes 70 images, and the testing dataset includes 70 images.

B. Experimental Environment

In this experiment, the operating system is Ubuntu 18.04, CPU is E5-2630, memory is 128G, GPU is NVIDIA TITAN Xp, GPU acceleration library is CUDA10.0. The deep learning framework uses Keras2.2.4 as the front end and TensorFlow1.13.1 as the back end.

C. Experimental Parameter Settings

1) Learning-rate: 0.001

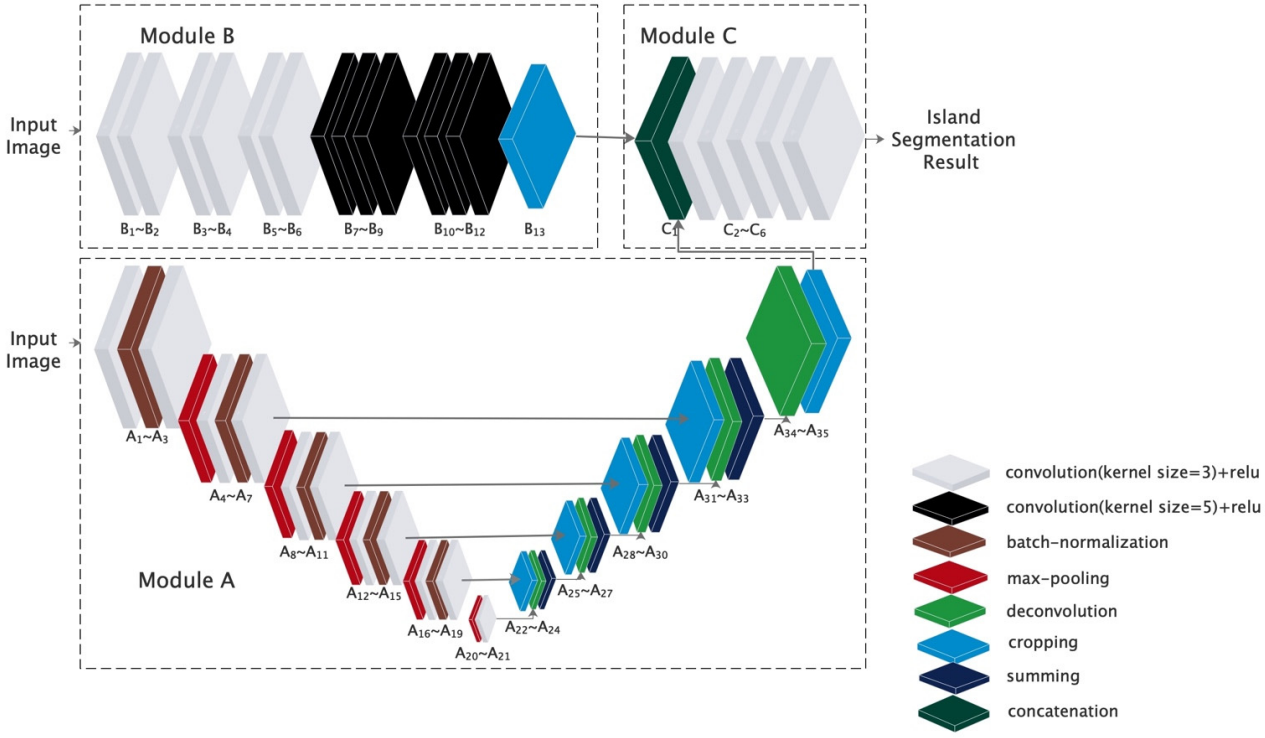


Figure 1. The network structure of FC_U-Net.

2) Weight decay coefficient: 0.0001;

3) Batch size: 6;

4) Epoch: 200;

5) Optimization method: Adam;

6) Loss function: cross-entropy loss function.

D. Evaluation Metrics

In this experiment, the mean accuracy[12], pixel accuracy, mean region intersection over union (mean IU), frequency weighted IU (fw IU), dice coefficient (DIC), and matthew correlation coefficient (MCC) are statistically calculated, whose computation equations are $mean\ accuracy = \frac{(\sum_c(n_{cc}/\sum_i n_{ci}))}{N}$, $pixel\ accuracy = \frac{\sum_c n_{cc}}{\sum_c \sum_i n_{ci}}$, $mean\ IU = \frac{\sum_c n_{cc}/(\sum_i n_{ic}+\sum_i n_{ci}-n_{cc})}{N}$, $fw\ IU = \frac{\sum_c((n_{cc}\times\sum_i n_{ci})/(\sum_i n_{ic}+\sum_i n_{ci}-n_{cc}))}{\sum_i\sum_c n_{ic}}$, $DIC = \frac{2\times TP}{2\times TP+FP+FN}$, $MCC = \frac{TP\times TN-FP\times FN}{\sqrt{(TP+FP)\times(TP+FN)\times(TN+FP)\times(TN+FN)}}$. respectively. The n_{ci} is the number of class i segmented to be class c, and $i, c \in \{island, none\ island\}$. The N represents the number of object category, and is set as 2 in this study for island and non-island object. The TP, FP, TN, and FN represent the numbers of true positives, false positives, true negatives, and false negatives, respectively.

E. Experimental Results and Analysis

In order to verify the effectiveness of the method in this paper, we designed three comparative experiments, respectively using three state-of-the-art methods, U-Net, FCN, SegNet. The

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same dataset, experimental parameters and evaluation standard with our experiment were used to ensure the reliability of the comparison results.

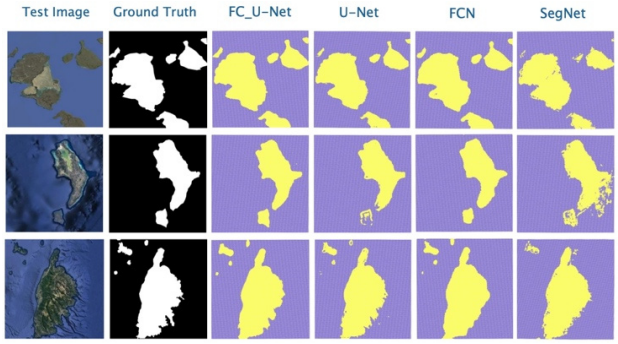


Figure 2. Visual effect comparison of test results of FC_U-Net, U-Net, FCN, SegNet.

The experimental results are shown in figure 2. From left to right, each column is the original images, ground truth, FC_U-Net detection results, U-Net detection results, FCN detection results and SegNet detection results. As can be seen from figure 2, U-Net can segment most of the small islands, but cannot segment an accurate island boundary. In addition, parts of the interior of the islands, such as the small island at the bottom of the second image, are incorrectly classified as the background. Some background interference is also generated, such as some noise at the bottom right of the third image. This is because[13] the semantic features of the islands are combined with some shallow details, which have a lot of interference. FCN can also segment most of the small islands, but the boundary of the segmented islands is relatively rough. For example, the small island at the bottom of the second image and the right side of the large island in the third image are quite different from the original images. This probably because[14] FCN uses the

Table 1. Quantitative evaluation results of FC_U-Net, U-Net, FCN, SegNet

Method	Mean Accuracy	Pixel Accuracy	Mean IU	Frequency Weighted IU	Mean DIC	MCC
FC_U-Net	0.8853	0.9045	0.8089	0.8255	0.8933	0.7887
U-Net	0.8153	0.8061	0.6650	0.6821	0.7973	0.6074
FCN	0.8221	0.8079	0.6691	0.6847	0.8005	0.6185
SegNet	0.8332	0.8645	0.7361	0.7601	0.8455	0.6982

feature information obtained from the upsampling operations, resulting in the loss of details. SegNet misses many small islands, such as the three islands at the top left of the third image. Besides, there is serious background interference. This probably because[15] it applies a little detail information from pooling indexes in segmentation. Compared with the above three networks, FC_U-Net can not only segment most of the small islands, but also can segment an accurate island boundary. Besides, the background interference is quite little. This is because the deepening of the network structure of Module A brings more semantic information so that small islands can be well segmented. Moreover, the addition of Module B removes the pooling layers and uses the deep convolutional layers to enrich detail information, avoid the reduction of detail information and the decrease of receptive field, making the boundary segmentation of the island more accurate and reducing the background interference.

In order to further verify the proposed viewpoint, we quantitatively evaluated the detection effects of each network, and the results are shown in table 1. The results show that FC_U-Net performs better than U-Net, FCN, SegNet in the mean accuracy, pixel accuracy, mean region intersection over union (mean IU), frequency weighted IU (fw IU), dice coefficient (DIC), and matthew correlation coefficient (MCC) ,which indicates the superiority of the network design.

Therefore, both visual and quantitative evaluation results show that our FC_U-Net performs better than U-Net, FCN and SegNet on this dataset.

F. Exploration and Analysis about The Function of Each Module

In order to study the role of each module in FC_U-Net, we trained three different networks for comparison. The first network is the whole FC_U-Net structure, the second network is Module A + Module C structure, the third network Module B + Module C structure. The same dataset, experimental parameters and evaluation standard with the previous experiments are used to ensure the reliability of the comparison results.

The experimental results are shown in figure 3. From left to right, each column is the original images, ground truth, the FC_U-Net detection results, the Module A+C detection results, and the Module B+C detection results. When Module B is removed, Module A+C can segment most of the small islands, but it cannot segment an exact island boundary. In addition, some parts of the interior of the island are wrongly classified as the background, such as the largest islands in the first, second and third images. In addition, some background interference is generated, especially in the third image. This is mainly because the semantic features of the islands are combined with some details of shallow layers, which have a lot of interference. When Module A is removed, Module B+C lose some small islands, such as the lost islands in the third image. This probably because the use of the convolution layers alone leads to insufficient semantic features and contextual information. After the fusion of Module A and Module B, the semantic information extracted by Module A and the detail information extracted by Module B are merged, so that not only can most of the small islands be

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segmented, but also the accurate island boundary can be segmented, and the background interference is quite little.

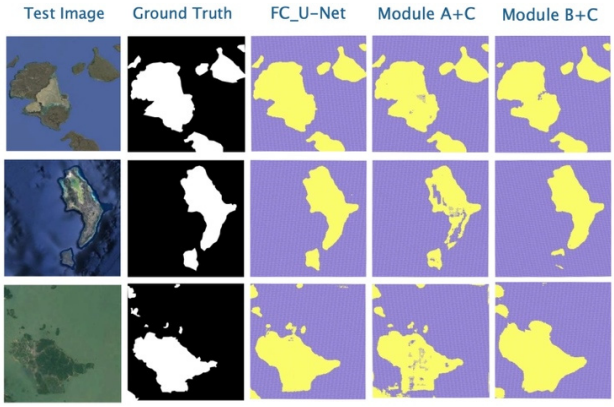


Figure 3. Visual effect comparison of test results of FC_U-Net, Module A+C, Module B+C.

In order to further verify the proposed viewpoint, we quantitatively evaluated the detection effects of the three networks, and the results are shown in table 2. The experimental results show that FC_U-Net performs better than Module A+C and Module B+C in the mean accuracy, pixel accuracy, mean region intersection over union (mean IU), frequency weighted IU (fw IU), dice coefficient (DIC), and matthew correlation coefficient (MCC) , which indicates that Module A and Module B play different roles in the whole network respectively, and neither of them is indispensable.

Therefore, both visual and quantitative evaluation results show that Module A and Module B both have a promoting effect on the network. They complement each other, neither of them is indispensable, so that the performance of the whole network can reach the best.

IV. CONCLUSION

Aiming at handling the problems of semantic segmentation for island remote sensing images, we propose a new network structure FC_U-Net, in which U-Net and FCN are improved respectively to form two new modules, and then a fusion module is used to fuse the semantic features and detail features from the two modules. After the trained network is verified and tested on the NWPU-RESISC45 dataset, both visual and quantitative evaluation results of FC_U-Net are better than other three state-of-the-art methods, U-Net, FCN, SegNet. In addition, we design and train two similar structures, which lack of corresponding network modules. The experimental results show that the two modules play substantial roles in the promotion of network performance. However, there are still some limitations of our work. For example, when the images are more complicated, the network segmentation accuracy is insufficient. And the network training is time-consuming. Next, we will focus on how to conduct semantic segmentation of remote sensing images more accurately and efficiently.

Table 2. Quantitative evaluation results of FC_U-Net, Module A+C, Module B+C

Method	Mean Accuracy	Pixel Accuracy	Mean IU	Frequency Weighted IU	Mean DIC	MCC
FC_U-Net	0.8853	0.9045	0.8089	0.8255	0.8933	0.7887
Module A+C	0.7837	0.8417	0.6835	0.7178	0.8063	0.6557
Module B+C	0.8067	0.8173	0.6736	0.6957	0.8026	0.6059

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